

Compositional monitoring of energy-mix drift on the simplex

A compositional change-detection protocol read natively in Aitchison geometry

CoDaWork 2026

11th International Workshop on Compositional Data Analysis

Coimbra, Portugal · 1–5 June 2026

P. Higgins · Rogue Wave Audio

with the HUF AI Collective · Claude · ChatGPT · Copilot · Gemini · Grok

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

Beat 1 — The simplex view

"Energy generation is a composition."

A country's electricity mix is a vector of carrier shares that sums to one.

Carriers (EMBER 9): coal · gas · oil · nuclear · hydro · wind · solar · bioenergy · other

Closure, perturbation, Aitchison distance, the simplex —
the geometry on which this entire talk operates.

Beat 2 — The data and the central claim

"No monitoring framework combines all three conjuncts."

Aitchison-native

Read in the simplex's own geometry — closure, CLR, ILR-Helmert. Not approximated; not lifted into a Euclidean ambient.

Formal change detection

Metric / pivot displacement quantified with TV distance and K-eff. Trajectory captured, not just snapshot.

Carrier-level attribution

The helmsman channel names which carrier moved. Structural change is decomposable.

Combined into one observable stack.

This conjunction is the falsifiable claim of the talk.

Beat 3 — The protocol

"Perturbation, Aitchison distance, TV distance, K-eff."

Perturbation

compositional shift

classical CoDa operator

Aitchison distance

the simplex metric

classical CoDa operator

TV distance

structural velocity

small supporting metric

K-eff

concentration measure

small supporting metric

INV-050 CANONICAL · TV distance / Aitchison distance pair-invariance verified across 101 datasets.

Self-discipline note: L2 → TV correction is on the record. We tell on ourselves when we mislabel.

Beat 4 — Japan

"Japan, Fukushima, 2011–2012, the spike."

Step-Δ Aitchison

× 3

over neighbouring years

Trajectory: abrupt external shock

Carrier-level attribution: nuclear → fossil substitution post-2011.

Helmsman flips: 17 (highest in the 8-country corpus).

Deceptive-drift transitions: 2.

Protocol verdict: regime change. Shock loud and unambiguous.

Beat 5 — Germany • the headline

"Germany, the trajectory, the deceptive drift, $p = 0.0016$ — with the null caveat."

p-value

0.0016

Trajectory: continuous arc toward the renewable vertex.

IR class: OVERDAMPED_EXTREME (smooth convergence).

Helmsman flips: 13.

Reading at monthly grain • the packet's headline result.

Null-model caveat — read out loud:

Computed against the series' own empirical-frequency baseline.

This is a weaker null than a Dirichlet, permutation, or bootstrap null.

Beat 6 — United Kingdom

"The UK coal exit registers as a regime change."

Japan

Abrupt · external

shock

UK

Abrupt · policy-driven

regime change

Helmsman flips: 15

Germany

Continuous · policy-driven

drift

The protocol discriminates among three archetypes of structural change.

Beat 7 — Beyond the three

"5 of 9 countries reproduce the deceptive-drift signature."

Signature fires (5)

AUS · Australia
CHN · China
GBR · United Kingdom
IND · India
JPN · Japan

Does not fire (4)

DEU · loud drift at annual grain
FRA · loosening regime
USA · stable composition
WLD · aggregate signal

INV-051 CANONICAL · the protocol discriminates · it does not over-fire.

Beat 8 — Three open questions for the room

"I do not have these answers. I am asking you."

Q1

Concentration vs Aitchison norm

Are K-eff (concentration) and Aitchison-norm tightening the same observable, or different facets?

Q2

Right family of simplex distances

Which family of valid simplex distances is appropriate for testing verdict-invariance across the trajectory? (INV-050 follow-up)

Q3

Right null model for compositional change-point

What null is appropriate for the deceptive-drift p-value? Dirichlet? Permutation? Bootstrap? (INV-051 follow-up — methodologically consequential)

Beat 9 — How to defeat this claim

"A defeater must combine all three conjuncts."

Metric defeat

show the verdict flips under a different valid simplex metric
· preempted by INV-050

Case defeat

show signature fires uniformly everywhere (no discrimination) · preempted by INV-051

Prior-art defeat

show the three-conjunct conjunction already exists · Morais et al. 2017/18 · Arata & Onozaki 2017 closest adjacent · 3 areas pending · open

Category defeat

show this is an application note within existing CoDa, not a new monitoring category · we have no preconceived answer · open

A defeater must combine all three conjuncts. Inspection is invited.

Beat 10 — Close

"Two repositories, one self-discipline note, thank you."

higgins-decomposition (Hs)

Deterministic engines · CNT v3.1.0 / CNQ v2.0.0

63 Investigation Catalog entries · Hs Change Control v1.0

Higgins-Unity-Framework (HUF)

MC-4 framing · EITT · HUF Governance Charter

KILL-001 (19 named failure modes) · case studies

Self-discipline note: L2 → TV correction is on the record.

Hashes carry the receipts. Anyone with the raw CSV can verify any plate in ~2 minutes.

The talk is an ascent waypoint, not the summit.

Thank you. I'll take questions.

Q&A bench · extended material on disk

"What sits behind the slides if the room asks."

EITT

Entropy-Invariant Time Transformer — Shannon entropy conserved under geometric-mean temporal compression;
0.18% / 341:1. papers/EITT_CANONICAL_EXPLANATION_2026-05-12.md

Twin-quaternion factoring

INV-029 CANONICAL · D=8 factoring verified at IEEE machine floor (3.33e-16) on EMBER China.

CHSH joint-coherence

INV-035 CANONICAL · 0.88 within Tsirelson bound.

KILL-001

19 named failure modes · published falsifiability artifact (HUF, 2026-03-23).

HUF Governance Charter

9 articles (April 2026) · parent doctrine for Hs Change Control v1.0.

Bread the Hs way

verbal pedagogy · the dinner-table version of the framework · papers/BREAD_THE_HS_WAY_2026-05-12.md

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